Chapter 18, Problem 67P

Problem

Given a signal $g(t)$ = sinc($200\pi\, t$), find the Nyquist rate and the Nyquist interval for the signal.

Step-by-step solution

Step 1 of 1

We are given,

$g(t) = \sin c(200\pi t)$

$so;\ \omega_c = 200\pi \Rightarrow f_c = \frac{\omega_c}{2\pi} = \frac{200\pi}{2\pi} = 100$

Nyquist rate is given by

$N_q \geq 2 f_c \qquad [f_m = \text{Maximum modulating frequency}]$

$or,\ N_q \geq 2 \times 100$

$N_q \geq 200 sampled / sec = \boxed{200 H_Z}$

Now, Nyquist interual for the signal.

$t \geq \frac{1}{N_q}$

$or,\ t \geq \frac{1}{200 H_Z}$

$or,\ t \geq 0.005 s = \boxed{5.0 ms}$